

ENGINEERING SPECIFICATIONS

Document Name: KAS/KWS/KAA Double Circuit Modbus List		
Document No: ESP-24-25-002	Revision: R03	Date:03-07-2024
Prepared By: PJP	Checked By: AS	Approved By: NMR

KAS/KWS/KAA Double Circuit				
Descriptions	UOM	Type	Read/Write	Modbus Add(Holding Register)
Evaporator Water In Temperature	°C	ANALOG	Read	40002
Evaporator Water Out Temperature	°C	ANALOG	Read	40003
LWT Setpoint	°C	ANALOG	Read/Write	40151
Compressor - 1				
Sump Oil Temperature	°C	ANALOG	Read	40004
Discharge Temperature	°C	ANALOG	Read	40005
Suction Temperature	°C	ANALOG	Read	40006
Liquid Line Temperature	°C	ANALOG	Read	40007
Condenser IN Temperature/Ambient	°C	ANALOG	Read	40008
Condenser Out Temperature	°C	ANALOG	Read	40009
Suction Pressure	Bar	ANALOG	Read	40010
Discharge Pressure	Bar	ANALOG	Read	40011
Oil Feed Pressure	Bar	ANALOG	Read	40012
Motor Current	Amp	ANALOG	Read	40013
Suction Superheat	°C	ANALOG	Read	40014
Percentage RLA	%	ANALOG	Read	40021
Evaporator Approach	-	ANALOG	Read	40022
Condenser Approach	-	ANALOG	Read	40032
Run Hours- High*	Hrs.	INTEGER	Read	45091
Run Hours- Low*	Hrs.	INTEGER	Read	45090
Number Of Starts H*	-	INTEGER	Read	45092
Number Of Starts L*	-	INTEGER	Read	45093
Compressor – 2				
Sump Oil Temperature	°C	ANALOG	Read	40034
Discharge Temperature	°C	ANALOG	Read	40035
Suction Temperature	°C	ANALOG	Read	40036
Liquid Line Temperature	°C	ANALOG	Read	40037
Condenser IN Temperature/Ambient	°C	ANALOG	Read	40038
Condenser Out Temperature	°C	ANALOG	Read	40039
Suction Pressure	Bar	ANALOG	Read	40040
Discharge Pressure	Bar	ANALOG	Read	40041
Oil Pressure	Bar	ANALOG	Read	40042
Motor Current	Amp	ANALOG	Read	40043
Suction Superheat	°C	ANALOG	Read	40044
Percentage RLA	%	ANALOG	Read	40051
Evaporator Approach	-	ANALOG	Read	40052
Condenser Approach	-	ANALOG	Read	40033
Run Hours- High*	Hrs.	INTEGER	Read	45089
Run Hours- Low*	Hrs.	INTEGER	Read	45088

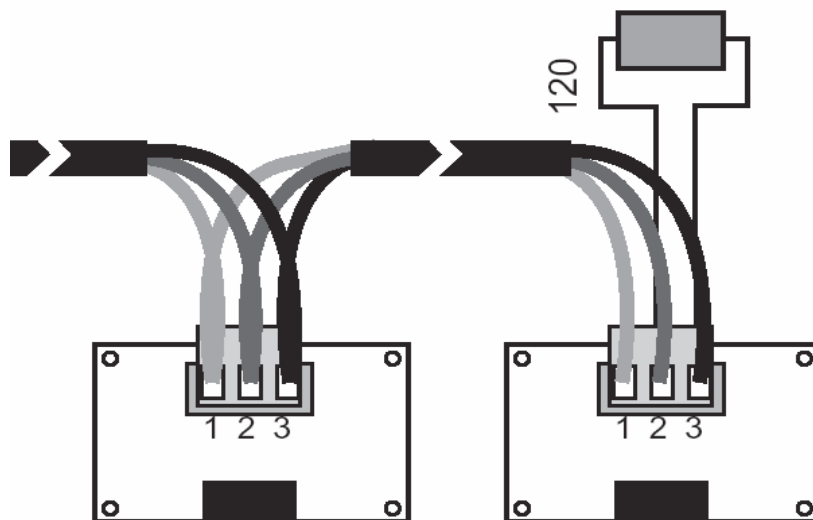
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Number Of Starts H*	-	INTEGER	Read	45094
Number Of Starts L*	-	INTEGER	Read	45095

Protocol Details	
Parity	None
Baud rate	19200
Protocol	Modbus/Modbus Extended
Start bit	1
Stop bit	1
No of bit	8
Communication Interface	RS-485
Interface Connection	2 Wire

The connection with the RS485 network is carried out using the plug-in terminal connector on the card. Pin-wiring of the connector is stamped on the card. If the card is placed in the last position of the supervision serial line, pins 2 and 3, you must connect a 120 Ω - 1/4 W end line resistor, as shown.



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pin	significato <i>meaning</i>
1	GND
2	RX+/TX+
3	RX-/TX-

*Total Run hours/No of starts will be calculated as below *Total run hrs = Run hour H (45023) x 1000 + Run hour L (45024) *Total no of starts = Number of starts H (45019) x 1000 + Number of starts L (45020)